

Sarvesh Bhattacharyya

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PROFILE SUMMARY

Final-year ECE student at MSRIT and AI & Backend Engineer with production experience in LLM-integrated products, retrieval pipelines, Voice AI systems. Built and shipped real AI systems across customer automation, multi-agent evaluation, knowledge graph retrieval, and robotics intelligence. Comfortable owning problems end-to-end from a blank slate. Focused on systems that work under real constraints: latency, cost, deployment, and reliability.

EXPERIENCE

AI Engineer Intern — Beaut Group (business.superowl.in)

Feb 2026 – Present · Bengaluru, Karnataka · On-site

- Independently designed and built the complete Voice AI automation layer for small-business customer interactions, owning the features codebase end-to-end across backend workflows, integration logic, and automated call-handling paths for multiple live business clients; reduced response latency from ~10s to ~3s.
- Engineered real-time communication, notification, and human-in-the-loop intervention flows, enabling business owners to stay aware of live Voice AI interactions and step in when required while routine conversations remained automated by default.
- Extended the system into a multi-tenant React Native owner-facing app with backend APIs, live call screens, missed-call summaries, persistent call logs, push notifications, and real-time updates, delivering all to production in under 3 months.

PROJECTS

ROAST — Live-Market Based Resume Critic [GitHub](#) | [Live Demo](#)

- ROAST pulls live market intelligence for your exact role, city, and company type before reading a single line — feedback calibrated against your actual competition this week, then gives critically honest feedback and fixes. Consists of a 6-agent pipeline including a TechnicalDepthAgent that web-searches unfamiliar technologies in real time before evaluating them, understanding what you built rather than just keyword-matching. Optional JD input parsed into structured requirements fed to all agents. 50+ analyses delivered at launch.
- Built the market intelligence store entirely offline — monthly ingestion pipeline into SQLite+FTS5+sqlite-vec (switched from hosted vector DB after discovering free tier suspends after 7 days). Reduced ingestion from 10 hours to 1 hour by verifying actual rate limits and tuning semaphore concurrency. DIVE retrieval: 6 targeted queries → parallel BM25 + vector → RRF fusion → distillation. Results stream via WebSocket; Redis-backed session state survives disconnects.
- ReviewAgent falls back across 7 LLM providers automatically with circuit breakers and proactive switching before rate limits hit. Quality gate enforces minimum output standards. Total monthly infrastructure cost: \$0.

Skills / Tools: FastAPI · asyncio · Redis · WebSocket · Groq · NVIDIA NIM · Langfuse · instructor · SQLite+FTS5 · React+Vite

SYNAPSE — Live AI Knowledge Graph Platform | [GitHub](#)

- Built a live, auto-updating relational knowledge graph for the AI ecosystem answering queries no flat feed can — 'which tools implement speculative decoding and support, sorted by stars' or 'what changed in RAG this month.' Ingests 11 free-tier sources daily via GitHub Actions; every relationship carries full provenance. Neo4j graph + Qdrant vector hybrid enables both structural and semantic search across 200K+ nodes.
- Safe NL-to-Cypher query layer (schema-constrained, write-blocked), multi-key LLM rotation across 3 Groq keys for 11× throughput, per-source circuit breakers, and domain-agnostic architecture — same core engine redeploys for aerospace, biotech, etc by swapping only a YAML config. \$0/month on free tier.

Skills / Tools: Neo4j Aura · Qdrant · PostgreSQL · LangGraph · Groq · FastAPI · React 19 · GitHub Actions

ACARE — Autonomous Clinical Assistance Robot | Institutionally Funded (₹40,000) · Owned entire software stack

- Built the complete software intelligence stack of a 6-DOF clinical robot letting clinic staff fetch instruments hands-free during long procedures. Owned 10+ ROS2 nodes, 10-state FSM coordinating voice/vision/motion/safety, dual biometric auth (voice d-vector + face embedding), dialogue management with context resolution, and full firmware integration protocol.
- Self-curated and annotated proprietary surgical instrument dataset from scratch; trained object detection model to 97.5% mAP. Bayesian NBV search learns tool placement patterns across sessions. Always-on ESTOP keyword thread (<200ms voice-to-halt, never mutable); graded 3-tier proximity safety via lidar; 3-check handover (face + palm + voice) before any tool release.

Skills / Tools: ROS2 · Python · C++ · YOLOv11 TFLite INT8 · LangGraph · Groq · Deepgram · SpeechBrain · MediaPipe · SQLite

SKILLS

Languages: Python, JavaScript, C++, Rust (basic)

Backend: FastAPI, REST APIs, WebSockets, asyncio, Redis, SQLite, PostgreSQL, Docker, GitHub Actions, AWS, Ubuntu

AI Engineering: LLM API Integration, Voice AI, RAG, Multi-Agent Systems, Vector Search, LangChain, LangGraph, Langfuse, Prompt Engineering

Data Systems: SQL, NoSQL, SQLite FTS5, Neo4j, Qdrant, ChromaDB, Pinecone, Data Pipelines, Retrieval Pipelines

Mobile / Edge AI: React Native, Firebase, YOLO, Deepgram

EDUCATION

MS Ramaiah Institute of Technology, Bengaluru

2022 – 2026

B.E. Electronics & Communication Engineering | CGPA: 7.47 / 10

Relevant Coursework: Data Structures, Operating Systems, Computer Networks, Python, C/C++

Machine Learning Specialization — DeepLearning.AI / Andrew Ng, Coursera

2026